

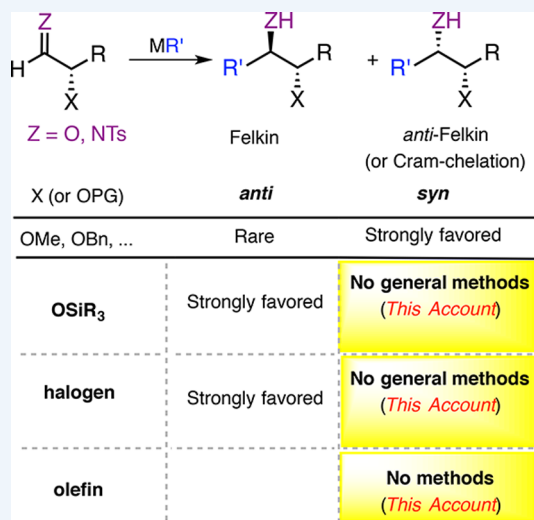
Chelation-Controlled Additions to Chiral α - and β -Silyloxy, α -Halo, and β -Vinyl Carbonyl Compounds

Xinyuan Fan[†] and Patrick J. Walsh^{*,†,‡,§}

[†]Institute of Advanced Synthesis, School of Chemistry and Molecular Engineering, Jiangsu National Synergetic Innovation Center for Advanced Materials, Nanjing Tech University, 30 South Puzhu Road, Nanjing 211816, P. R. China

[‡]Roy and Diana Vagelos Laboratories, Department of Chemistry, University of Pennsylvania, 231 South 34th Street, Philadelphia, Pennsylvania 19104-6323, United States

CONSPECTUS: The science and art of preventing and managing disease and prolonging life is dependent on advances in medicine, biology, and biochemistry. Many of these advances will involve interactions of small molecules with biological entities. As such, they will rely on the efficient synthesis of active compounds with very high stereochemical purity. Although enantioselective reactions are important in this regard, most stereocenters in complex molecule synthesis are installed in diastereoselective reactions. Perhaps the most well-known diastereoselective C–C bond-forming reaction is the addition of nucleophiles to carbonyl groups with α - or β -stereogenic centers. Diastereoselective additions of *organometallic reagents* to protected chiral α - and β -hydroxy aldehydes and ketones are described by either Cram chelation or Felkin–Anh models, which are protecting group (PG)-dependent. Small PGs (X = OMOM, OBn, etc.) favor *Cram chelation*, wherein *both the carbonyl group and the O-PG* bind to the Lewis acidic metal, providing *syn diol* motifs. In contrast, silyl PGs, with the OSiR₃ moiety being both bulky and weakly coordinating, provide *anti diols* (Felkin addition). It is well-known that exceptions to this paradigm are scarce. Therefore, the choice of PG is based on the desired stereochemical outcome in the addition step and is often inappropriate for the global protection strategy. Thus, it is critical to develop general methods for chelation-controlled additions of organometallics to chiral silyloxy aldehydes and ketones. Once the challenge of developing chelation-controlled additions to silyloxy carbonyl compounds can be met, the next question is what other pendant functional groups can chelate?



Herein we introduce the first general methods for the chelation-controlled addition of organometallics to chiral silyloxy aldehydes and ketones. A wide variety of organozinc reagents have been used in these addition reactions, including dialkylzinc reagents that are commercially available or generated using Knochel's methods. Existing protocols for the generation of (*E*)-di- and -trisubstituted vinylzinc reagents have been employed, and new methods for the generation of (*Z*)-di- and -trisubstituted vinylzinc reagents have been developed. The generation of 1,1-heterobimetallic reagents based on boron and zinc has been advanced, and the addition of these reagents to silyloxy aldehydes via chelation-control is included. We will first describe the initial discovery and a model to explain the observed diastereoselectivities. A wide array of chelation-controlled additions to chiral α - and β -silyloxy aldehydes and ketones will then be presented.

We next describe other functional groups that undergo chelation-controlled additions. α -Halo aldehyde derivatives are well-known to favor Felkin addition (via the Cornforth–Evans model). We introduce a general method for chelation-controlled additions to α -halo aldimines that provides useful precursors to aziridines. Finally, we provide preliminary evidence that even C=C bonds can play the role of chelating groups in additions to β,γ -unsaturated ketones.

The results outlined in this Account redefine the commonly held idea that chiral silyloxy- and halo-substituted carbonyl compounds only give Felkin addition products. The key to achieving chelation control in these reactions is the use of weakly coordinating solvents (dichloromethane and toluene) that do not readily bind to the zinc Lewis acids RZnX.

1. INTRODUCTION

The total synthesis of natural products generally starts with a member of nature's pool of enantioenriched compounds or the application of an early-stage enantioselective transformation.¹ Whatever the origin of the enantioenriched precursor, the

prevailing strategy for the synthesis is the exploitation of the substrate's existing stereochemistry for the controlled formation

Received: June 25, 2017

Published: August 15, 2017

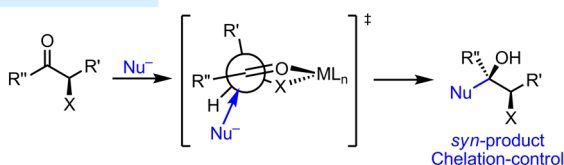


of new bonds and stereocenters. Among diastereoselective-substrate-controlled reactions, the addition of organometallics to C=O bonds situated α or β to stereocenters is of great practical importance.

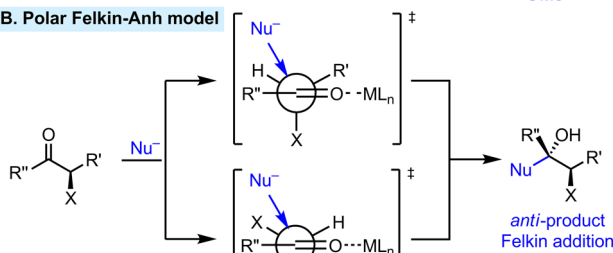
Groundbreaking studies by Cram described methods to predict and control the diastereoselectivity in nucleophilic additions to C=O bonds with neighboring stereocenters.^{2,3} Fundamental contributions from Cornforth,⁴ Felkin,⁵ Anh and Eisenstein,⁶ and Evans,⁷ among others,^{8–11} have collectively formed the foundation of current stereoinduction models.^{10,12,13} On the basis of these models (Scheme 1), the

Scheme 1. Models for Stereoinduction

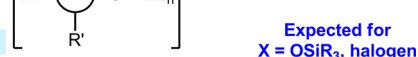
A. Cram-chelation model



B. Polar Felkin-Anh model



C. Cornforth-Evans model

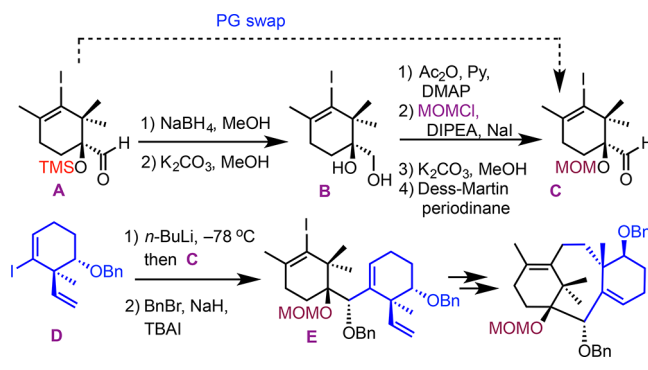


outcomes of nucleophilic additions of organometallics to protected α - or β -hydroxy carbonyl compounds is protecting group (PG)-dependent. Thus, small PGs (methyl, benzyl, etc.), where the oxygen can easily coordinate to Lewis acids, undergo carbonyl additions via the Cram chelation pathway (chelation control; Scheme 1A).^{3,12} Sterically demanding PGs, such as SiR_3 , disfavor coordination to metals, and additions proceed through the polar Felkin–Anh pathway and result in Felkin products (Scheme 1B).^{6,12,14} Although exceptions to these models are rare,^{8,12,15–19} examples of binding of silyl ethers to metals have been known for many years.^{20,21} Important contributions to chelation-controlled additions with silyl ether protecting groups were made by the groups of Reetz,¹⁰ Evans,¹⁵ and Eliel,^{8,17} to name a few. The third important model is the Cornforth–Evans model (Scheme 1C),^{7,22} which is most relevant to α -halogenated carbonyl derivatives.¹² The Cornforth–Evans model minimizes electrostatic interactions (dipole moments) through an anti conformation of the carbonyl and C–X bonds (Scheme 1C). While the underlying principles of the Cornforth–Evans and polar Felkin–Anh models are different, the products have the same stereochemistry (Felkin products). The models depicted in Scheme 1 are outlined in advanced organic textbooks and discussed in graduate organic courses.

These powerful models enable accurate predictions of the stereochemical outcomes of organometallic additions to chiral carbonyl compounds. The drawback of this paradigm, however, is that the protecting group strategy is dictated by the product

stereochemistry: *anti*-diols require α -silyloxy carbonyls while vicinal *syn*-diols employ α -hydroxy carbonyls with PGs that facilitate coordination (MOM, Bn, etc.). The issue is best illustrated with an example (Scheme 2). The TMS group of A

Scheme 2. Cost of PG Swapping



was removed, followed by reprotection with MOMCl to give C. Lithiation of D followed by chelation-controlled addition to C gave E.²³ Less laborious methods to control the diastereoselectivity include the use of enantioenriched catalysts (catalyst control).²⁴

Scheme 2 illustrates a long-standing problem: methods for highly stereoselective chelation-controlled additions to silyloxy carbonyl compounds are lacking. Herein we outline our efforts to develop chelation-controlled additions of organozinc reagents to substrates that are well-known to resist chelation (α - and β -silyloxy carbonyls, α -chloro imines, and even β,γ -unsaturated ketones).

2. ADDITIONS OF (Z)-VINYLZINC REAGENTS TO CHIRAL ALDEHYDES

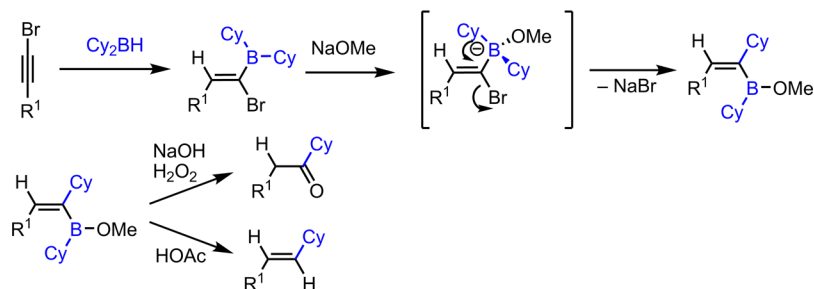
Our interests date back to the synthesis of stereodefined vinylzinc reagents and their additions to aldehydes and ketones to prepare allylic alcohols.^{25,26} Inspiration for the preparation of (Z)-vinylzinc intermediates originated from Zweifel,²⁷ who discovered that 1-halo-1-alkynes undergo highly regioselective hydroborations with dicyclohexylborane (Scheme 3). 1-Halo-1-alkenylboranes formed in this way were subjected to addition of nucleophiles, such as sodium methoxide, alkylolithiums,²⁸ Grignards,²⁸ and hydrides.²⁹ These reagents deliver the nucleophile to boron, which then undergoes a 1,2-metalate rearrangement with one of the boron-bound groups migrating to the vinyl center (Scheme 3A). The rearrangement is stereospecific, resulting in inversion at the vinyl center. We envisioned that transmetalation of (Z)-vinylboranes with dialkylzincs^{30,31} would provide reactive (Z)-vinylzinc reagents.

In the synthesis of (Z)-disubstituted vinylzinc reagents (Scheme 3B), hydroboration of 1-bromoalkynes with HBCy_2 led to 1-bromo vinylboranes as single regioisomers. Addition of *t*-BuLi and warming the reaction mixture from -78°C to room temperature led to the formation of the (Z)-vinylborane. Addition of diethylzinc initiated the transmetalation to generate the (Z)-vinylzinc reagent, which smoothly added to a variety of achiral aldehydes (>80% yield; Scheme 3B).³² An enantioselective version was also introduced.³³ Addition to α - and β -benzyloxy aldehydes led to chelation-controlled addition products (11:1 to 15:1 dr; Scheme 3C).³²

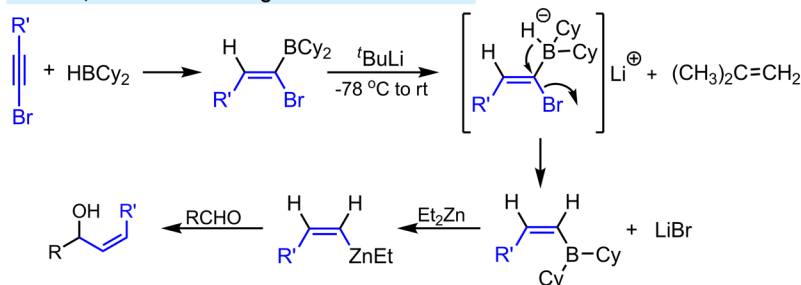
Additions to α -TBS-protected hydroxy aldehydes led to the formation of allylic alcohols with 4:1 dr. Lewis acids generally

Scheme 3. (A) Zweifel's 1,2-Metallate Rearrangement Followed by Oxidation or Protodeboration; (B) Transmetalation/Addition Strategy; (C) Chelation-Controlled Addition with PG = Bn

A. Zweifel's 1,2-metallate rearrangement



B. Our 1,2-metallate rearrangement/transmetalation



C. Addition to benzyloxy aldehydes

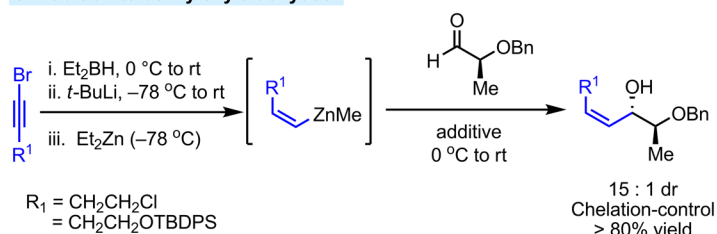


Table 1. Diastereoselective Additions of (Z)-Disubstituted Vinylzinc Reagents to Silyloxy Aldehydes

entry	bromoalkyne	chiral aldehyde	major product (chelation-control)	additive	dr (anti:syn)	yield (%)
1				$\text{BF}_3 \cdot \text{OEt}_2$	8 : 1	70
2				$\text{BF}_3 \cdot \text{OEt}_2$	8 : 1	82
3				–	18 : 1	60
4				ZnBr_2	6 : 1	89
5				ZnBr_2	6 : 1	84

increase the diastereoselectivity in carbonyl additions. Thus, the use of $\text{BF}_3 \cdot \text{OEt}_2$ increased the dr to 8:1 (Table 1, entries 1 and

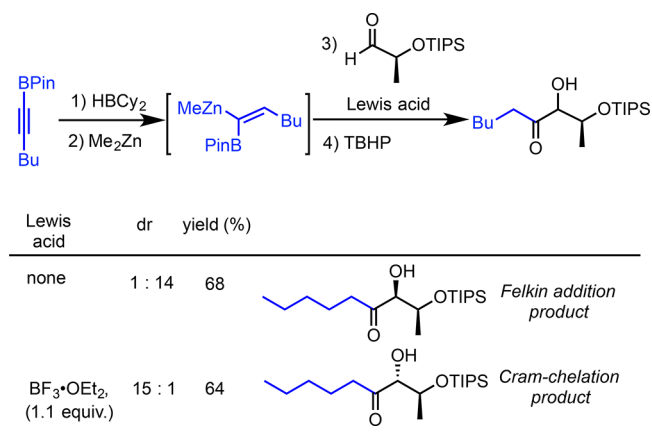
2). Throughout this work, stereochemical assignments were performed by X-ray analysis, ^1H NMR analysis of Mosher

esters,³⁴ or conversion to known compounds. We were surprised (and confused!) to find the products in entries 1 and 2 were derived from *chelation control*. This is particularly odd because $\text{BF}_3 \cdot \text{OEt}_2$ is a *monodentate* Lewis acid, which normally favors Felkin addition. The *O*-benzyl-containing alkyne underwent chelation-controlled addition with 18:1 dr in the absence of additive (entry 3). The more challenging TBS-protected β -hydroxy aldehyde exhibited chelation control in the presence of ZnBr_2 (6:1 dr; entries 4 and 5).

The results of our first foray into the Felkin versus Cram additions left more questions than answers, but it was clear that the stereochemical outcomes were synthetically valuable and stood in stark contrast to the paradigm that silyloxy aldehydes do not undergo chelation-controlled additions.

We also generated boron/zinc heterobimetallics using an analogous strategy (Scheme 4). We were able to control the diastereoselectivity in additions even to TIPS-protected aldehydes when using $\text{BF}_3 \cdot \text{OEt}_2$.³⁵

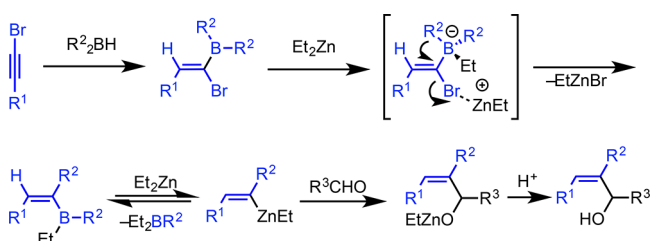
Scheme 4. Control of Diastereoselectivity with Zn/B Heterobimetallics



Following the logic in Scheme 3B, we wondered whether organozincs would add to boron and induce the 1,2-rearrangement to afford (*Z*)-trisubstituted vinylboranes and ultimately (*Z*)-trisubstituted vinylzinc reagents (Scheme 5). In Scheme 5, however, either R^2 or Et can migrate. Migration selectivities as high as 14:1 were observed ($\text{R}^2 = \text{Cy}$), but in most cases the migration selectivity was $\sim 4:1$.

As outlined in Scheme 5, hydroboration of 1-bromoalkynes with diethyl- or dicyclohexylborane was followed by treatment with diethylzinc at -78°C to induce rearrangement and transmetalation. Addition to achiral aldehydes proceeded in 60–84% yield. Reactions with chiral α - and β -hydroxy aldehydes with benzyl PGs exhibited chelation control (7:1

Scheme 5. Generation of (*Z*)-Trisubstituted Vinylzinc Reagents and Additions to Aldehydes



to 20:1).³⁶ The TBS-protected α -hydroxy aldehyde underwent addition with a variety of trisubstituted vinylzincs to form chelation-controlled products ($>20:1$ dr; Table 2). Even the

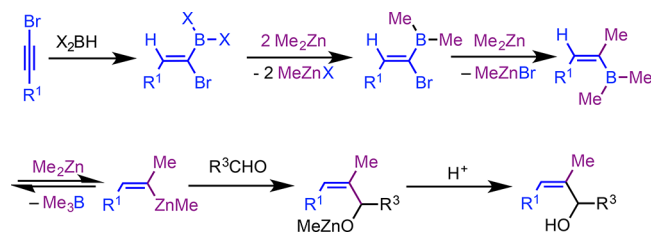
Table 2. Diastereoselective Additions of (*Z*)-Trisubstituted Vinylzincs to Silyloxy Aldehydes

entry	R^1	R^2	RCHO	yield (%)	dr (anti:syn)
1	<i>n</i> -Bu	Et		82	$>20:1$
2	$\text{Cl}(\text{CH}_2)_3$	Et		92	$>20:1$
3	$\text{TBDPSO}(\text{CH}_2)_2$	Et	$\text{H}-\text{C}(=\text{O})-\text{CH}(\text{OTBS})-\text{Me}$	95	$>20:1$
4	<i>n</i> -Bu	Cy	$\text{H}-\text{C}(=\text{O})-\text{CH}(\text{OTBS})-\text{Me}$	75	$>20:1$
5	$\text{Cl}(\text{CH}_2)_3$	Cy		54	$>20:1$
6	$\text{TBDPSO}(\text{CH}_2)_2$	Cy		60	$>20:1$
7	<i>n</i> -Bu	Et	$\text{H}-\text{C}(=\text{O})-\text{CH}(\text{OTIPS})-\text{Me}$	74	$>20:1$

TIPS-protected α -hydroxy aldehyde gave $>20:1$ diastereoselectivity. In the presence of an enantioenriched amino alcohol-based catalyst, trisubstituted allylic alcohols could be prepared with enantioselectivities up to 97%.³⁶

To make the methyl-substituted (*Z*)-trisubstituted vinylzincs, pyrophoric gaseous HBMe_2 is required. To circumvent this limitation, the hydroboration was performed with HBX_2 ($\text{X} = \text{Cl}, \text{Br}$; Scheme 6). Addition of Me_2Zn initiates substitution of the B–X bonds and then induces the 1,2-metalate rearrangement and transmetalation, followed by addition to the aldehyde.

Scheme 6. Approach to α -Methyl (*Z*)-Trisubstituted Substituted Allylic Alcohols



The α -methyl (*Z*)-trisubstituted vinylzincs generated in Scheme 6 ($\text{X} = \text{Br}$) were added to a variety of achiral aldehydes (70–92% yield).³⁷ Additions of α -methyl (*Z*)-trisubstituted vinylzincs to chiral α - or β -hydroxy aldehydes possessing PGs that traditionally chelate resulted in chelation control (9:1 to $>20:1$ dr, 47–70% yield).³⁶ As outlined in Table 3, hydroboration with HBCl_2 followed by addition of Me_2Zn and enantioenriched aldehyde provided the α -methyl (*Z*)-trisubstituted allylic alcohols in 67–78% yield. For comparison, dimethylborane was prepared and used in the vinylation reactions. In these cases, the yields were higher (75–78%;

Table 3. Synthesis of α -Methyl (Z)-Trisubstituted Allylic Alcohols

entry	R ¹	R ²	yield (%)	dr (anti:syn)
1	<i>n</i> -Bu	Cl	68	>20:1
2	Cl(CH ₂) ₃	Cl	67	>20:1
3	TBDPSO(CH ₂) ₂	Cl	78	>20:1
4	<i>n</i> -Bu	Me	78	>20:1
5	TBDPSO(CH ₂) ₂	Me	75	>20:1

entries 4 and 5). In all cases, the diastereomeric ratio was >20:1, again favoring chelation-controlled additions.

In this section, a straightforward and stereospecific approach to the formation of (Z)-di- and -trisubstituted vinylzinc intermediates has been introduced. These reagents add to α - and β -siloxy-protected aldehydes with excellent stereochemical control. Remarkably, the resulting products are derived from *chelation-controlled* additions. Our hypothesis is that the RZnX byproducts in Scheme 5 and 6 behave as chelating Lewis acids. In reactions with BF₃·OEt₂, the active Lewis acid is likely RZnF or RZn⁺.

3. HIGHLY DIASTERESELECTIVE CHELATION-CONTROLLED ADDITIONS TO α - AND β -SILYLOXY ALDEHYDES AND KETONES

On the basis of the hypothesis that the active Lewis acids in these additions are RZnX (X = Cl, Br; R = Me, Et), we prepared RZnX complexes (R = Et, Me, *n*-Bu) by conproportionation of ZnR₂ with ZnX₂.³⁸ To probe our hypothesis, we examined the reaction of Et₂Zn with silyloxy aldehydes in the absence and presence of different amounts of EtZnCl and EtZnBr (Table 4). TBS-protected α -hydroxypropanal showed no diastereoselectivity in the absence of EtZnCl. Increasing the amount of EtZnCl from 0.25 to 1.5 equiv (with respect to aldehyde) resulted in an increase in the dr to ~30:1 favoring the chelation-controlled addition. A similar trend was observed with EtZnBr. Likewise, the TES- and TIPS-protected substrates reached dr's of >20:1 with 1.0–1.5 equiv of EtZnCl (the reaction with the TIPS derivative was conducted at –50 °C to increase the diastereoselectivity). Furthermore, combining Et₂Zn and TBS-protected α -hydroxypropanal without EtZnCl resulted in no product after 8 h at –50 °C. In contrast, with 1.5 equiv of EtZnCl, 65% conversion was observed in 16 min. An increase in reaction rate is anticipated if the reaction proceeds via a chelated intermediate.^{17,39}

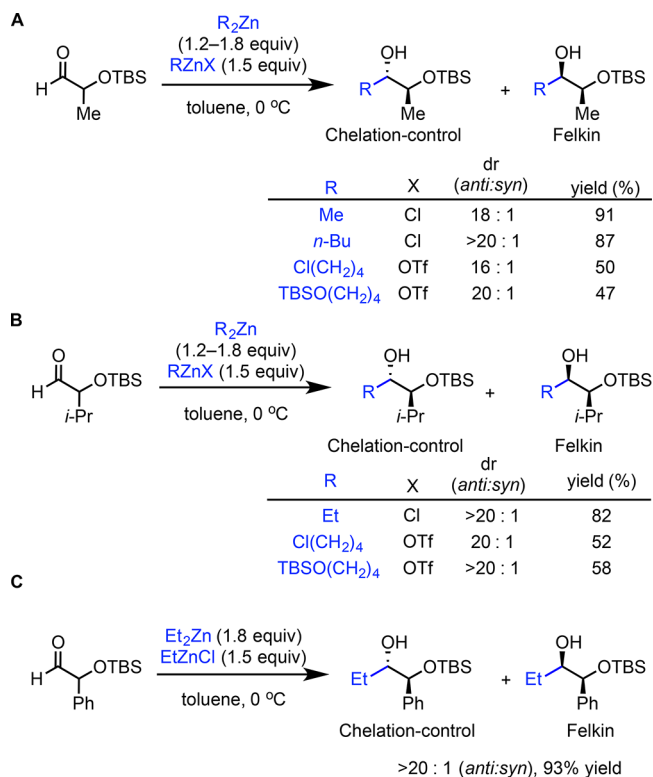
An examination of other zinc reagents was next conducted. It is known that alkyl exchange among zinc species is rapid,⁴⁰ so the alkyl group on the Lewis acid RZnX must be the same as

Table 4. Diastereoselective Additions to Silyl-Protected α -Hydroxypropanals

entry	PG	EtZnX	equiv	yield (%)	dr (anti:syn)
1	TBS	EtZnCl	0	77	1:1
2			0.25	91	6.7:1
3			0.5	94	14:1
4			1.0	93	31:1
5			1.5	90	28:1
6	TBS	EtZnBr	0.10	81	6:1
7			0.25	89	13:1
8			1.0	94	20:1
9			1.5	92	22:1
10	TES	EtZnCl	0	69	1:1
11			0.50	72	15.4:1
12			1.0	85	>20:1
13			1.5	77	>20:1
14	TIPS	EtZnCl	0	60	1:2
15			1.5	80	>20:1

those on R₂Zn. As shown in Scheme 7, the use of Me₂Zn with MeZnCl or (*n*-Bu)₂Zn with *n*-BuZnCl and TBS-protected α -

Scheme 7. Chelation-Controlled Alkyl Additions to Racemic Aldehydes

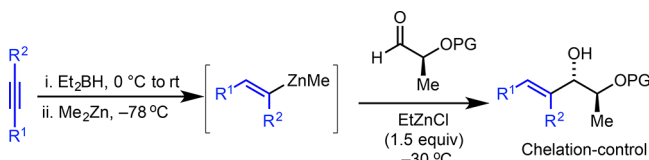


hydroxypropanal formed the chelation-controlled products with $\geq 18:1$ dr. Knochel and Jones⁴¹ have reported excellent methods to prepare functionalized organozinc reagents, which we employed here. These organozinc reagents (2.7 equiv with respect to the aldehyde) were treated with 1.5 equiv of HOTf to generate the mixture of R₂Zn and the Lewis acid RZn(OTf)

for use in the additions in Scheme 7A. Chelation-controlled addition products were isolated from these reactions with high diastereoselectivities. The lower yields arise from aldehyde reduction, likely through β -hydrogen transfer. Additional aldehydes (Scheme 7B,C) also provided chelation-controlled products with high dr. The high diastereoselectivities in Scheme 7 suggest that this approach should be appropriate to a host of related aldehydes. Further studies led to development of an autocatalytic version of this reaction,⁴² which is covered elsewhere.

Given the importance of stereochemically defined allylic alcohols, we expanded our diastereoselective additions to (*E*)-vinylzinc reagents. Hydroboration of terminal alkynes with Et_2BH followed by in situ boron to zinc transmetalation^{30,31} with Et_2Zn generates the vinylzinc intermediates. In the presence of EtZnCl , addition to TBS-protected (*S*)- α -hydroxypropanals takes place at 0 °C, giving good yields (54–93%) of allylic alcohols with diastereoselectivities $\geq 10:1$ favoring chelation control (Table 5). The enyne in entry 6

Table 5. Generation of (*E*)-Di- and -Trisubstituted Allylic Alcohols



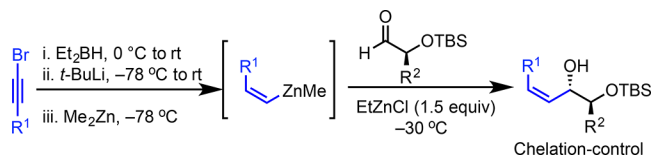
entry	PG	alkyne	yield (%)	dr (anti:syn)
1	TBS	Ph-C≡C-H	82	>20 : 1
2	TBS	<i>n</i> -Bu-C≡C-H	54	17 : 1
3	TBS	 -C≡C-H	82	11 : 1
4	TBS	TBDPSO(CH ₂) ₂ -C≡C-H	72	>20 : 1
5	TBS	Cl(CH ₂) ₂ -C≡C-H	92	10 : 1
6	TBS	 -C≡C-H	70	>20 : 1
7	TBS	<i>t</i> -Bu-C≡C-Me	85	>20 : 1
8	TES	<i>t</i> -Bu-C≡C-Me	93	>20 : 1

afforded the *anti*-dienyl alcohol in 70% yield with >20:1 dr. The unsymmetrical internal alkyne *tert*-butyl methyl acetylene was also an excellent substrate with either TBS- or TES-protected (*S*)- α -hydroxypropanal, providing the chelation-controlled addition products in high yields with >20:1 dr. It is noteworthy that this method is scalable—the reaction in entry 1 was conducted on a 5 mmol scale.

Given the high diastereoselectivities in the (*E*)-vinylations with EtZnCl (Table 5), we reexamined the (*Z*)-vinylations outlined in Scheme 3B and Table 1. As previously outlined, in the absence of a Lewis acid additive, generation of the (*Z*)-disubstituted vinylzinc and addition to TBS-protected α -hydroxypropanal afforded the product with 4:1 dr. Use of $\text{BF}_3\cdot\text{OEt}_2$ increased the diastereoselectivity to 8:1 (Table 1, entries 1 and 2). The use of EtZnCl as the Lewis acid and

addition of these same vinylzinc reagents to TBS-protected α -hydroxypropanal resulted in $\geq 14:1$ dr (Table 6, entries 1 and

Table 6. Revisitation of Diastereoselective (*Z*)-Vinylations of Silyloxy Aldehydes

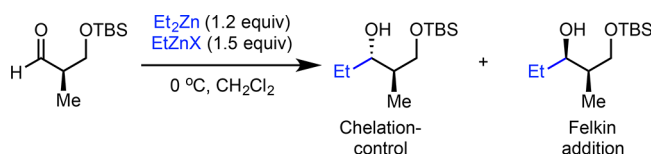


entry	alkyne	R ²	yield (%)	dr (anti:syn)
1	TBDPSO-C≡C-Br	Me	78	14 : 1
2	Cl-C≡C-Br	Me	72	20 : 1
3	TBDPSO-C≡C-Br	<i>i</i> Pr	80	>20 : 1
4	Cl-C≡C-Br	<i>i</i> Pr	90	>20 : 1

2). Use of the *i*-Pr-substituted aldehyde led to >20:1 dr (entries 3 and 4). The advantage of RZnX is that they are much less aggressive and more selective than $\text{BF}_3\cdot\text{OEt}_2$.

The broad scope and excellent diastereoselectivities in the EtZnCl -promoted addition of alkyl and vinylzinc reagents to α -silyloxy aldehydes led us to wonder whether similar selectivities would be possible with β -silyloxy aldehydes, which are more challenging substrates than their α -silyloxy counterparts because they form larger chelates with Lewis acids (entropic effects).⁴³ In the case of the TBS α -silyloxy aldehyde, addition of Et_2Zn with 1.5 equiv of EtZnCl generated the product with 28:1 dr (Table 4, entry 5), compared with $\sim 9:1$ dr using the TBS β -silyloxy aldehyde (Scheme 8). Although 9:1 is synthetically useful, we screened other Lewis acids and found EtZnOTf to perform better (15.5:1 dr).

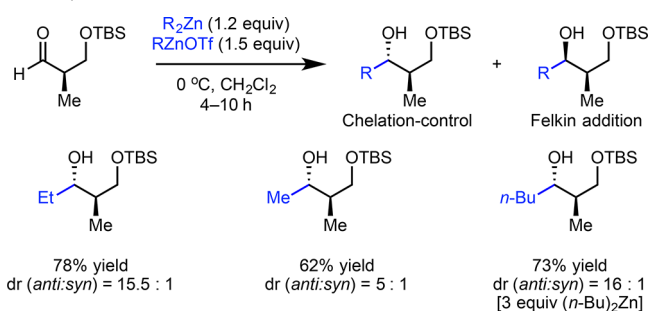
Scheme 8. Lewis Acid Screen with β -Silyloxy Aldehyde



EtZnX	dr (anti:syn)	yield (%)
EtZnCl	8.9 : 1	89
EtZnBr	7.9 : 1	85
EtZnOTf	15.5 : 1	78
EtZnONf	10 : 1	87
$\text{EtZn}(\text{TF}_2)$	6 : 1	72

In the presence of MeZnOTf , methyl addition proceeded with only 5:1 diastereoselectivity favoring chelation control (Scheme 9). This was partially attributed to the reduced solubility of MeZnOTf in dichloromethane. Butyl addition with $(n\text{-Bu})_2\text{Zn}/(n\text{-Bu})\text{ZnOTf}$ required 3 equiv of $(n\text{-Bu})_2\text{Zn}$ to minimize the reduction product. Ultimately the chelation-controlled product was isolated in 73% yield with 16:1 dr. This

Scheme 9. Alkyl Additions to TBS-Protected β -Silyloxy Aldehydes



approach was used by Morken in the total synthesis of (+)-discodermolide.⁴⁴

Vinylations of protected β -silyloxy aldehydes required extensive optimizations, leading to the use of a 1:1 toluene/ CH_2Cl_2 mixture as the solvent system at $-15\text{ }^\circ\text{C}$. The use of EtZnOTf in entry 1 of Table 7 resulted in 4.8:1 dr, possibly as a

Table 7. Vinyl Additions to Protected β -Silyloxy Aldehydes

entry	PG	alkyne	yield (%)	dr (anti:syn)
1	TBS	<i>n</i> -Bu—C≡C—H	61	10 : 1
2	TBS	<i>t</i> -Bu—C≡C—H	70	10 : 1
3	TBS		66	5.8 : 1
4	TBS	<i>t</i> -Bu—C≡C—Me	80	20 : 1
5	TBS	<i>i</i> -Pr—C≡C—Me	78	17 : 1
6	TES	<i>n</i> -Bu—C≡C—H	65	11 : 1
7	TES		78	9 : 1
8	TES	<i>t</i> -Bu—C≡C—H	71	18 : 1
9	TES	<i>t</i> -Bu—C≡C—Me	76	> 20 : 1

result of decreased solubility of EtZnOTf under the reaction conditions. In contrast, the diastereoselectivity was 10:1 with EtZnONf . As shown in entries 1–3 and 6–8 of Table 7, (*E*)-disubstituted allylic alcohols were generated in 61–80% yield with dr $\geq 5.8:1$. (*E*)-trisubstituted allylic alcohols were accessible with $\geq 17:1$ dr in 76–80% yield (entries 4, 5, and 9). It is noteworthy that TES-protected 3-hydroxy-2-methylpropanal generally underwent additions with higher diastereoselectivity. The smaller size of the TES group relative to the TBS group likely renders chelation more favorable.

To this point, the focus has been on aldehydes. However, there are numerous natural products that possess vicinal secondary and tertiary diols. We therefore focused on extending the diastereoselective additions to α -silyloxy ketones.⁴⁵ At the outset, we were concerned that the reduced reactivity of ketones, compounded with the mild nature of organozinc reagents and the moderate Lewis acidity of RZnX compounds, would result in little or no addition and instead would lead to enolate formation.

Consistent with these concerns, treatment of TBS-protected 3-hydroxybutanone with Et_2Zn in the absence of Lewis acids resulted in $\sim 5\%$ conversion after several days at $0\text{ }^\circ\text{C}$ with no diastereoselectivity. Application of 1.5 equiv of EtZnX led to addition products contaminated with aldol byproducts. On the basis of optimization experiments, the use of 3.5 equiv of RZnCl provided the best conversions and excellent diastereoselectivities (Table 8). This is illustrated by additions of

Table 8. Alkyl Additions to Methyl Ketones

entry	ketone	ZnR_2	yield (%)	dr (anti:syn)
1		ZnEt_2	91	>20 : 1
2		ZnBu_2	86	>20 : 1
3		ZnEt_2	96	>20 : 1
4		ZnBu_2	78	>20 : 1
5		ZnEt_2	88	>20 : 1
6		ZnBu_2	81	>20 : 1
7		ZnEt_2	85	>20 : 1
8		ZnBu_2	78	>20 : 1
9		ZnEt_2	68	>20 : 1

Et_2Zn (1.5 equiv) and $(n\text{-Bu})_2\text{Zn}$ (3.5 equiv) to a variety of methyl ketones protected with TBS or TES groups (Table 8). When less $(n\text{-Bu})_2\text{Zn}$ was employed, more aldol products formed. Excellent diastereoselectivities were observed with ketones bearing α -methyl, α -phenyl, and α -benzyl groups and with TBS or TES PGs.

Replacement of the α' -methyl substituent of the ketones in Table 8 with *n*-butyl resulted in a surprising decrease in reactivity. Unlike α' -methyl ketones, which reacted at $-15\text{ }^\circ\text{C}$ (Table 8), the α' -*n*-Bu ketones underwent additions at $50\text{ }^\circ\text{C}$ in the presence of 3.5 equiv of EtZnCl and 4 equiv of Et_2Zn . As shown in Table 9, addition of Et_2Zn to TBS-protected 2-hydroxyheptan-3-one occurred with excellent diastereoselectivity (>20:1, chelation control) but in moderate yield (54%). Me_2Zn , which exhibits much lower reactivity than Et_2Zn , added to TBS-protected 2-hydroxyheptan-3-one to generate the

Table 9. Alkyl Additions to Butyl Ketones

entry	ketone	ZnR ₂	yield (%)	dr (anti:syn)
1		ZnEt ₂	54	>20 : 1
2		ZnMe ₂	81	>20 : 1
3		ZnEt ₂	51	>20 : 1
4		ZnMe ₂	86	>20 : 1
5		ZnEt ₂	56	>20 : 1
6		ZnMe ₂	77	>20 : 1

product in 81% yield with >20:1 dr. The TES-protected analogue reacted with diethyl- and dimethylzinc to afford chelation-controlled products in 51–86% yield with >20:1 dr (entries 3 and 4). The TBS-protected α -phenyl ketones reacted with Et₂Zn and Me₂Zn in similar yields and diastereoselectivities (entries 5 and 6).

Terminal alkynes (2–2.5 equiv) were employed in the vinylation of ketones via hydroboration/transmetalation to Me₂Zn (Table 10). Additions to both TBS- and TES-protected 3-hydroxybutanone at –30 °C with 3 equiv of EtZnCl generated tertiary (*E*)-disubstituted allylic alcohols with dr $\geq$ 18:1 favoring chelation control (entries 1–3 and 6–8). These

Table 10. Diastereoselective Additions of (*E*)-Di- and -Trisubstituted Vinylzinc Reagents

entry	PG	alkyne	yield (%)	dr (anti:syn)
1	TBS		88	>20 : 1
2	TBS		81	>20 : 1
3	TBS		83	18 : 1
4	TBS		78	>20 : 1
5	TBS		65	>20 : 1
6	TES		91	>20 : 1
7	TES		68	>20 : 1
8	TES		64	>20 : 1
9	TES		81	>20 : 1
10	TES		80	>20 : 1

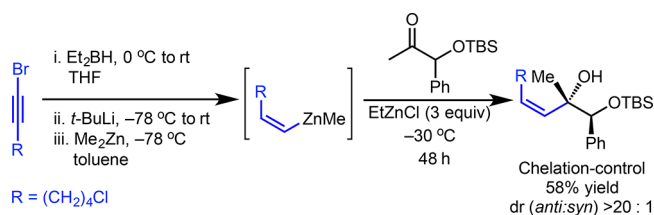
results include the dienyl alcohol with 18:1 dr in 83% yield. It is also noteworthy that the use of 5 mmol of cyclopropylacetylene led to the allylic alcohol product with >20:1 dr in 81% yield (entry 2). Internal alkynes proved to be viable in additions to ketones, providing access to trisubstituted allylic alcohols in 65–91% yield with >20:1 dr (entries 4, 5, 9, and 10).

We also examined α -silyloxy ketones bearing an α -phenyl substituent (Table 11). Despite the increased acidity of the benzylic α -C–H, additions proceeded in 58–83% yield with >20:1 diastereoselectivity with both terminal and internal alkynes.

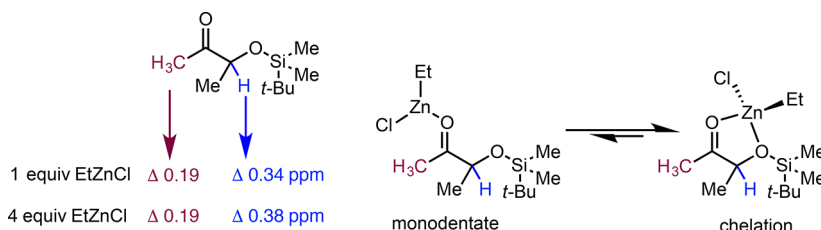
Table 11. Diastereoselective Vinylation of α -Phenyl Ketones

entry	PG	alkyne	yield (%)	dr (anti:syn)
1	TES		71	>20 : 1
2	TES		83	>20 : 1
3	TES		63	>20 : 1
4	TBS		74	>20 : 1
5	TBS		76	>20 : 1
6	TBS		58	>20 : 1

We were curious whether more sterically congested (*Z*)-vinylzincs would add to ketones. Generation of the (*Z*)-vinylzinc from the 1-bromo-1-alkyne was performed as outlined in Scheme 3B. Because of the strong binding of ethereal solvents to RZnX, the THF solvent had to be removed and replaced with a noncoordinating solvent (toluene). The (*Z*)-vinylborane was then subjected to Me₂Zn, EtZnCl, and the α -silyloxy ketone (Scheme 10), affording the (*Z*)-allylic alcohol in 58% yield with >20:1 dr.⁴⁵

Scheme 10. Diastereoselective (*Z*)-Vinylation of an α -Silyloxy Ketone

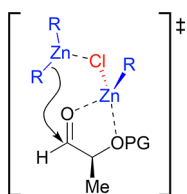
Given the predominance of chelation-controlled products in these organozinc additions promoted by RZnX Lewis acids, we were curious whether spectroscopic evidence for chelation could be gained. Thus, a ¹H NMR spectrum of TBS-protected 3-hydroxybutanone was acquired in CD₂Cl₂. Next, 1 equiv of EtZnCl was added, and the spectrum was reacquired. The methine hydrogen shifted downfield by Δ 0.34 ppm, while the

Scheme 11. ^1H NMR Evidence for Chelation of the Ketone

α' -methyl moved only Δ 0.19 ppm (Scheme 11). Since zinc complexes are weak Lewis acids, it is likely that the equilibrium lies toward the free ketone. An additional 3 equiv of EtZnCl was added, and another ^1H NMR spectrum was taken. The downfield shift of the methine (Δ 0.38 ppm) doubled that for the α' -methyl group (Δ 0.19 ppm). Further, upon addition of the Lewis acid, the SiMe_2 groups went from an apparent singlet to two distinct peaks. These data are most consistent with chelation of the ketone to EtZnCl (Scheme 11).

Upon inspection of the tables of substrates above, it is apparent that the diastereoselectivities of the silyloxy carbonyl compounds are exceptionally high and similar to what might be expected for traditional chelating groups such as benzyl. To account for the excellent control over stereochemical induction, we proposed a highly ordered transition state wherein RZnX acts both as a Lewis acid to chelate the substrate and as a Lewis base by forming a bridging halide interaction with R_2Zn (Scheme 12).⁴⁶ Coordination of RZnCl by R_2Zn increases the Lewis acidity of RZnCl while simultaneously increasing the ability of R_2Zn to transfer the alkyl group.

Scheme 12. Proposed Transition State



4. CHELATION-CONTROLLED ADDITIONS TO α -CHLORO ALDIMINES

The remarkable diastereoselectivity observed in the chelation-controlled additions to silyloxy aldehydes and ketones led us to wonder what other Lewis basic groups might chelate. Of course, alkyl halides are significantly less Lewis basic than silyl ethers. The prevailing wisdom would lead one to predict that α -halo carbonyl derivatives would undergo addition of nucleophiles via the Cornforth–Evans pathway, which minimizes dipole interactions (Scheme 1C). On the other hand, coordination of alkyl halides to transition metal centers has been known for some time.⁴⁷ Examples of chelation-controlled additions to α -halo carbonyl or imino substrates all involve hydride nucleophiles with α -fluoro carbonyl derivatives.^{48–50} With these ideas in mind, we set out to examine additions of organozinc reagents to α -halo carbonyl compounds.⁵¹ Our initial work with α -halo aldehydes gave very high diastereoselectivities favoring chelation control, but aldehyde reduction products dominated. This prompted us to examine imine derivatives.

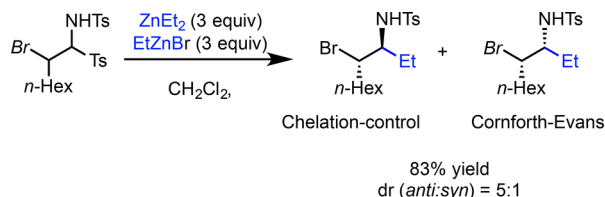
Because N -sulfonyl imines are difficult to handle, imine precursors were employed. Our goal was to develop a one-pot procedure wherein excess Et_2Zn deprotonates the precursor and generates the reactive α -chloro aldimine.⁵² However the optimization proved challenging because, along with control of the diastereoselectivity, significant reduction of the imine to the amine was observed with diethylzinc. Ultimately, we found that 3 equiv each of EtZnBr and Et_2Zn gave the best results at 0°C in DCM with an addition:reduction product ratio of 6.8:1 and 11:1 dr favoring chelation control. As shown in Table 12, a variety of α -chloro aldimines underwent addition with Et_2Zn and Me_2Zn to afford chelation-controlled products (5:1 to >20:1 dr).

Table 12. Addition of Et_2Zn and Me_2Zn to α -Chloro Imines

entry	aldimine precursor	ZnR_2	yield (%)	dr (anti:syn)
1		ZnEt_2	80	11 : 1
2		ZnMe_2	83	10 : 1
3		ZnEt_2	84	>20 : 1
4		ZnMe_2	80	>20 : 1
5		ZnEt_2	76	12 : 1
6		ZnMe_2	82	20 : 1
7		ZnEt_2	69	7 : 1
8		ZnMe_2	85	6 : 1
9		ZnEt_2	60	5 : 1
10		ZnMe_2	68	11 : 1

For the purpose of comparison to α -chloro aldimines in Table 12, an α -bromo aldimine was examined under similar conditions (Scheme 13). In the event, Et_2Zn added to the α -bromo aldimine with 5:1 diastereoselectivity (83% yield, chelation control). This result shows that chelation-controlled addition is not limited to α -chloro imines.

The synthesis of functionalized allylic amines is of great interest, which compelled us to examine the vinylation of α -chloro aldimines. Optimal yields and diastereoselectivities were

Scheme 13. Addition to an α -Bromo Aldimine

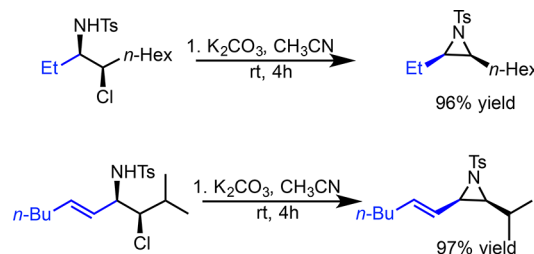
obtained by combining the in situ-formed vinylborane reagent with Me_2Zn at -78°C followed by EtZnBr and the aldimine precursor. The solution was then allowed to warm to -45°C . As outlined in entries 1–7 of Table 13, both di- and

Table 13. Vinylation of Aldimines

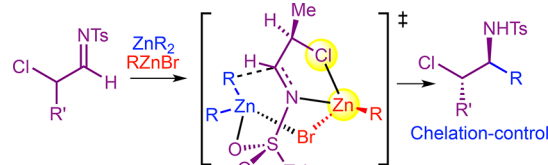
entry	precursor	alkyne	yield (%)	dr (<i>anti:syn</i>)
1		$n\text{-Bu}\text{---}\text{C}\equiv\text{C}\text{---}\text{H}$	89	>20 : 1
2		$\text{Ph}\text{---}\text{C}\equiv\text{C}\text{---}\text{H}$	72	>20 : 1
3		$t\text{-Bu}\text{---}\text{C}\equiv\text{C}\text{---}\text{H}$	72	>20 : 1
4		$\text{Cl}(\text{CH}_2)_3\text{---}\text{C}\equiv\text{C}\text{---}\text{H}$	83	>20 : 1
5			70	>20 : 1
6		$i\text{-Pr}\text{---}\text{C}\equiv\text{C}\text{---}\text{Me}$	70	>20 : 1
7		$\text{Me}\text{---}\text{C}\equiv\text{C}\text{---}\text{Me}$	69	>20 : 1
8		$n\text{-Bu}\text{---}\text{C}\equiv\text{C}\text{---}\text{H}$	79	>20 : 1
9		$n\text{-Bu}\text{---}\text{C}\equiv\text{C}\text{---}\text{H}$	77	>20 : 1
10		$n\text{-Bu}\text{---}\text{C}\equiv\text{C}\text{---}\text{H}$	81	20 : 1
11		$n\text{-Bu}\text{---}\text{C}\equiv\text{C}\text{---}\text{H}$	61	>20 : 1

trisubstituted vinylzincs add readily, providing access to allylic β -chloroamines with $\geq 20:1$ dr favoring chelation control. Likewise, an array of aldimine precursors were employed ($\geq 20:1$ dr; entries 8–11). It is also noteworthy that these reactions are practical: a 2 g scale reaction generated the chelation-controlled product with $>20:1$ dr in 89% yield. Furthermore, an enantioenriched aldimine precursor did not result in erosion of ee during the vinylation (entry 8). Finally, the β -chloroamine products are easily transformed into aziridines by treatment with K_2CO_3 ($>95\%$ yield; Scheme 14).

In collaboration with Prof. Per-Ola Norrby, we calculated several transition states for the addition reactions. Briefly, it was found that in the lowest-energy transition state structures, RZnBr binds the Cl and N of the aldimine while a bridging bromide and sulfonyl oxygen bind R_2Zn (Scheme 15).

Scheme 14. Conversion of β -Chloroamines to Aziridines

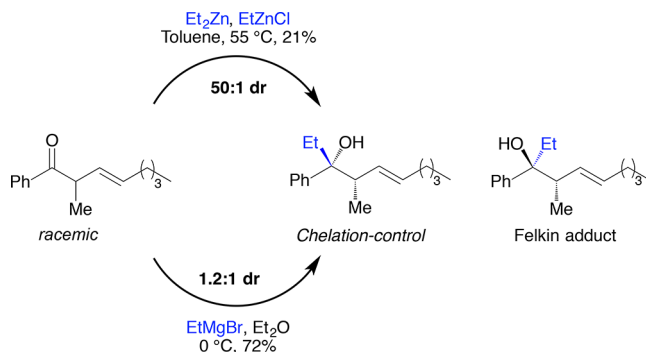
Scheme 15. Drawing of the Calculated Lowest-Energy Transition State



5. ADDITIONS TO β,γ -UNSATURATED KETONES

We again pondered what other groups on the substrate might enable chelation-controlled additions mediated by organozincs. We previously reported the first structure of a zinc–olefin complex.⁵³ Although $\text{Zn}(\text{II})$ is a d^{10} species, like $\text{Pd}(0)$, the d electrons of $\text{Zn}(\text{II})$ are quite low in energy and unable to significantly participate in π back-bonding. Thus, it was not surprising that our zinc–olefin complex was stable only in the solid state—dissolution led to dissociation. Nonetheless, very weak interactions are sufficient to control reaction outcomes. In fact, we had previously proposed weak Zn–arene interactions to rationalize a reversal in diastereoselectivity in Simmons–Smith halocyclopropanations.^{25,54}

Thus, we initiated investigations to determine whether β,γ -unsaturated ketones would chelate to zinc via the carbonyl and alkene moieties. The reaction optimization was plagued with problems due to the predominance of aldol products under all conditions examined. Nonetheless, as shown in Scheme 16,

Scheme 16. Ethyl Additions to a β,γ -Enone

addition of the enone to EtZnCl (2 equiv) and ZnEt_2 (3 equiv) in toluene at 55°C provided the chelation-controlled adduct with 50:1 dr (as determined by GC). Unfortunately, the yield was only 21%. By way of comparison, the use of EtMgBr showed almost no selectivity (1.2:1 dr). The remarkable selectivity with $\text{Et}_2\text{Zn}/\text{EtZnCl}$ provides support for our hypothesis that the reaction takes place via chelation control. Further evidence was gained through computational work

performed in collaboration with Prof. Marisa Kozlowski and Osvaldo Gutierrez and correlated with solution ^1H NMR binding studies.⁵⁵

7. CONCLUSIONS

We have clearly demonstrated that the range of coordinating groups for chelation-controlled carbonyl additions and related processes extends beyond small PGs and includes silyloxy groups, halogens, and double bonds. These studies have redefined the conventional thinking in chelation-controlled carbonyl additions. *Key to the observed selectivities is conducting the reactions in toluene and dichloromethane*, which do not bind well to RZnX Lewis acids, leaving chelation to the silyloxy and carbonyl oxygens as the best option to satisfy its coordination sphere.

AUTHOR INFORMATION

Corresponding Author

*E-mail: pwalsh@sas.upenn.edu.

ORCID

Patrick J. Walsh: 0000-0001-8392-4150

Notes

The authors declare no competing financial interest.

Biographies

Xinyuan Fan received his B.A. and M.A. (2010) from Lanzhou University in China and his Ph.D. (2014) at the Institute of Chemical Research of Catalonia in Spain with Prof. Miquel Pericas, with an exchange experience at the University of Pennsylvania with Prof. Patrick J. Walsh. He did postdoctoral research (2015–2017) with Prof. Peng Chen and Prof. Zhang-Jie Shi at Peking University, after which he joined Nanjing Tech University as an associate professor. He is developing organic methods and bioorthogonal chemical tools.

Patrick J. Walsh received his B.A. from UCSD and his Ph.D. at UC Berkeley with Prof. Robert G. Bergman (1991), and did postdoctoral research with Prof. K. B. Sharpless at Scripps. He then was an assistant professor at San Diego State University (1994–1999) and a professor at Centro de Graduados e Investigación, Instituto Tecnológico de Tijuana (1996–1999). In 1999 he moved to the University of Pennsylvania, where he is the Alan G. MacDiarmid Professor. He is also an Honorary Professor at Nanjing Tech University.

ACKNOWLEDGMENTS

We acknowledge our co-workers, without whose experimental skills and intellectual input this work would not have been possible. We are grateful to the NSF (CHE-1464744) for support.

REFERENCES

- (1) Walsh, P. J.; Kozlowski, M. C. *Fundamentals of Asymmetric Catalysis*; University Science Books: Sausalito, CA, 2008.
- (2) Cram, D. J.; Elhafez, F. A. A. Studies in stereochemistry. X. The rule of "steric control of asymmetric induction" in the synthesis of acyclic systems. *J. Am. Chem. Soc.* **1952**, *74*, 5828–5835.
- (3) Cram, D. J.; Kopecky, K. R. Studies in Stereochemistry. XXX. Models for Steric Control of Asymmetric Induction. *J. Am. Chem. Soc.* **1959**, *81*, 2748–2755.
- (4) Cornforth, J. W.; Cornforth, R. H.; Mathew, K. K. 24. A General Stereoselective Synthesis of Olefines. *J. Chem. Soc.* **1959**, 112–127.
- (5) Chérest, M.; Felkin, H.; Prudent, N. Torsional strain involving partial bonds. The stereochemistry of the lithium aluminum hydride

reduction of some simple open-chain ketones. *Tetrahedron Lett.* **1968**, *9*, 2199–2204.

(6) Anh, N. T.; Eisenstein, O. Theoretical Interpretation of 1–2 Asymmetric Induction- Importance of Anti-Periplanarity. *Nouv. J. Chim.* **1977**, *1*, 61–70.

(7) Evans, D. A.; Siska, S. J.; Cee, V. J. Resurrecting the Cornforth model for carbonyl additions: studies on the origin of 1,2-asymmetric induction in enolate additions to heteroatom-substituted aldehydes. *Angew. Chem., Int. Ed.* **2003**, *42*, 1761–1765.

(8) Chen, X.; Hortelano, E. R.; Eliel, E. L.; Frye, S. V. Are Chelates Truly Intermediates in Cram's Chelate Rule? *J. Am. Chem. Soc.* **1990**, *112*, 6130–6131.

(9) Karabatsos, G. J. Asymmetric Induction. A model for additions to carbonyls directly bonded to asymmetric carbons. *J. Am. Chem. Soc.* **1967**, *89*, 1367–1371.

(10) Reetz, M. T. Structural, mechanistic, and theoretical aspects of chelation-controlled carbonyl addition reactions. *Acc. Chem. Res.* **1993**, *26*, 462–468.

(11) Lodge, E. P.; Heathcock, C. H. Acyclic Stereoselection. 40. Steric Effects, as well as sigma*-Orbital Energies, are Important in Diastereoface Differentiation in Additions to Chiral Aldehydes. *J. Am. Chem. Soc.* **1987**, *109*, 3353–3361.

(12) Mengel, A.; Reiser, O. Around and Beyond Cram's Rule. *Chem. Rev.* **1999**, *99*, 1191–1224.

(13) Smith, R. J.; Trzoss, M.; Bühl, M.; Bienz, S. The Cram Rule Revisited Once More - Revision of the Felkin–Anh Model. *Eur. J. Org. Chem.* **2002**, *2002*, 2770–2775.

(14) Guillarme, S.; Ple, K.; Banchet, A.; Liard, A.; Haudrechy, A. Alkynylation of Chiral Aldehydes: Alkoxy-, Amino-, and Thio-substituted Aldehydes. *Chem. Rev.* **2006**, *106*, 2355–2403.

(15) Evans, D. A.; Allison, B. D.; Yang, M. G.; Masse, C. E. The Exceptional Chelating Ability of Dimethylaluminum Chloride and Methylaluminum Dichloride. The Merged Stereochemical Impact of alpha- and beta-Stereocenters in Chelate-Controlled Carbonyl Addition Reactions with Enolsilane and Hydride Nucleophiles. *J. Am. Chem. Soc.* **2001**, *123*, 10840–10852.

(16) Reetz, M. T.; Huellmann, M. Non-Chelation-Controlled Nucleophilic Addition to Chiral alpha-Alkoxy Ketones. *J. Chem. Soc., Chem. Commun.* **1986**, 1600–1602.

(17) Chen, X.; Hortelano, E. R.; Eliel, E. L.; Frye, S. V. Chelates as Intermediates in Nucleophilic Additions to Alkoxy Ketones According to Cram's Rule (Cyclic Model). *J. Am. Chem. Soc.* **1992**, *114*, 1778–1784.

(18) Evans, D. A.; Allison, B. D.; Yang, M. G. Chelate-Controlled Carbonyl Addition Reactions. The Exceptional Chelating Ability of Dimethylaluminum Chloride and Methylaluminum Dichloride. *Tetrahedron Lett.* **1999**, *40*, 4457–4460.

(19) Evans, D. A.; Halstead, D. P.; Allison, B. D. Chelation-Controlled Stannylacetylene Additions to β -Alkoxy Aldehydes Promoted by Alkylaluminum Halide Lewis Acids. *Tetrahedron Lett.* **1999**, *40*, 4461–4462.

(20) Hattori, K.; Yamamoto, H. Highly selective generation and application of (E)- and (Z)-silyl ketene acetals from α -hydroxy esters. *J. Org. Chem.* **1993**, *58*, 5301–5303.

(21) Williard, P. G.; Hintze, M. J. The first structural characterization of a dimeric lithium ketone enolate-LDA complex. *J. Am. Chem. Soc.* **1987**, *109*, 5539–5541.

(22) Cee, V. J.; Cramer, C. J.; Evans, D. A. Theoretical Investigation of Enolborane Addition to α -Heteroatom-Substituted Aldehydes. Relevance of the Cornforth and Polar Felkin–Anh Models for Asymmetric Induction. *J. Am. Chem. Soc.* **2006**, *128*, 2920–2930.

(23) Kawada, H.; Iwamoto, M.; Utsugi, M.; Miyano, M.; Nakada, M. Synthetic Studies on the Taxane Skeleton: Construction of Eight-Membered Carbocyclic Rings by the Intramolecular B-Alkyl Suzuki–Miyaura Cross-Coupling Reaction. *Org. Lett.* **2004**, *6*, 4491–4494.

(24) Soai, K.; Shimada, C.; Takeuchi, M.; Itabashi, M. Chiral Catalyst Controlled Addition of Dialkylzincs to Chiral α -Siloxyaldehydes; Asymmetric Synthesis of either Diastereoisomer of Mono-Protected Vicinal Diols. *J. Chem. Soc., Chem. Commun.* **1994**, 567–578.

- (25) Kim, H. Y.; Walsh, P. J. Efficient Approaches to the Stereoselective Synthesis of Cyclopropyl Alcohols. *Acc. Chem. Res.* **2012**, *45*, 1533–1547.
- (26) Hussain, M. M.; Walsh, P. J. Tandem Reactions for Streamlining Synthesis: Enantio- and Diastereoselective One-pot Generation of Functionalized Epoxy Alcohols. *Acc. Chem. Res.* **2008**, *41*, 883–893.
- (27) Zweifel, G.; Arzoumanian, H. α -Halovinylboranes. Their Preparation and Conversion into cis-Vinyl Halides, trans-Olefins, Ketones, and trans-Vinylboranes. *J. Am. Chem. Soc.* **1967**, *89*, 5086–5088.
- (28) Brown, H. C.; Imai, T.; Bhat, N. G. Vinylic Organoboranes. 7. Stereoselective Synthesis of (E)-(1-Substituted-1-alkenyl)boronic Esters by the Nucleophilic Substitution of (Z)-(1-Bromo-1-alkenyl)-boronic Esters with Organolithium or Grignard Reagents. Isolation and Oxidation to Ketones. *J. Org. Chem.* **1986**, *51*, 5277–5282.
- (29) Campbell, J. B. J.; Molander, G. A. Improved Synthesis of cis-Alkenylboranes. *J. Organomet. Chem.* **1978**, *156*, 71–79.
- (30) Srebnik, M. Stereospecific Preparation of Trisubstituted Allylic Alcohols by Alkene Transfer from Boron to Zinc. *Tetrahedron Lett.* **1991**, *32*, 2449–2452.
- (31) Oppolzer, W.; Radinov, R. N. Catalytic Asymmetric Synthesis of Secondary (E)-Allyl Alcohols from Acetylenes and Aldehydes via (1-Alkenyl)zinc Intermediates. *Helv. Chim. Acta* **1992**, *75*, 170–173.
- (32) Jeon, S.-J.; Fisher, E. L.; Carroll, P. J.; Walsh, P. J. Direct, Stereospecific Generation of (Z)-Disubstituted Allylic Alcohols. *J. Am. Chem. Soc.* **2006**, *128*, 9618–9619.
- (33) Salvi, L.; Jeon, S.-J.; Fisher, E. L.; Carroll, P. J.; Walsh, P. J. Catalytic Asymmetric Generation of (Z)-Disubstituted Allylic Alcohols. *J. Am. Chem. Soc.* **2007**, *129*, 16119–16125.
- (34) Ohtani, I.; Kusumi, T.; Kashman, Y.; Kakisawa, H. High-Field FT NMR Application of Mosher's Method. The Absolute Configurations of Marine Terpenoids. *J. Am. Chem. Soc.* **1991**, *113*, 4092–4096.
- (35) Li, H.; Carroll, P. J.; Walsh, P. J. Generation and Tandem Reactions of 1-Alkenyl-1,1-Heterobimetallals: Practical and Versatile Reagents for Organic Synthesis. *J. Am. Chem. Soc.* **2008**, *130*, 3521–3531.
- (36) Kerrigan, M. H.; Jeon, S.-J.; Chen, Y.; Salvi, L.; Carroll, P. J.; Walsh, P. J. (Z)-Trisubstituted Vinylzinc Reagents: Catalytic Asymmetric Addition to Prochiral Aldehydes and Diastereoselective Additions to Chiral Aldehydes. A One-pot Procedure. *J. Am. Chem. Soc.* **2009**, *131*, 8434–8445.
- (37) Chen, Y. K.; Walsh, P. J. A One-Pot Multicomponent Coupling Reaction for the Stereocontrolled Synthesis of (Z)-Trisubstituted Allylic Alcohols. *J. Am. Chem. Soc.* **2004**, *126*, 3702–3703.
- (38) Fabicon, R. M.; Richey, H. G. Formation of Coordinated $RZn(\text{macrocycle})^+$ Cations and Organozincate Anions from Reactions of Organozinc Compounds and Macrocycles. *Organometallics* **2001**, *20*, 4018–4023.
- (39) Frye, S. V.; Eliel, E. L.; Cloux, R. Rapid-Injection Nuclear Magnetic Resonance Investigation of the Reactivity of α - and β -Alkoxy Ketones with Dimethylmagnesium: Kinetic Evidence for Chelation. *J. Am. Chem. Soc.* **1987**, *109*, 1862–1863.
- (40) Blake, A. J.; Shannon, J.; Stephens, J. C.; Woodward, S. Demonstration of Promoted Zinc Schlenk Equilibria, their Equilibrium Values and Derived Reactivity. *Chem. - Eur. J.* **2007**, *13*, 2462–2472.
- (41) Knochel, P.; Jones, P. *Organozinc Reagents*; Oxford University Press: Oxford, U.K., 1999.
- (42) Raffier, L.; Stanton, G. R.; Walsh, P. J. Chelation-Controlled Additions to α -Silyloxy Aldehydes: An Autocatalytic Approach. *Org. Lett.* **2013**, *15*, 6174–6177.
- (43) Carey, F. A.; Sundberg, R. J. *Advanced Organic Chemistry: Part A: Structure and Mechanisms*, 5th ed.; Springer: New York, 2007; p 179.
- (44) Yu, Z.; Ely, R. J.; Morken, J. P. Synthesis of (+)-Discodermolide by Catalytic Stereoselective Borylation Reactions. *Angew. Chem., Int. Ed.* **2014**, *53*, 9632–9636.
- (45) Stanton, G. R.; Koz, G.; Walsh, P. J. Highly Diastereoselective Chelation-Controlled Additions to α -Silyloxy Ketones. *J. Am. Chem. Soc.* **2011**, *133*, 7969–7976.
- (46) Stanton, G. R.; Johnson, C. N.; Walsh, P. J. Overriding Felkin Control: A General Method for Highly Diastereoselective anti-Felkin Additions to α -Silyloxy Aldehydes. *J. Am. Chem. Soc.* **2010**, *132*, 4399–4408.
- (47) Kulawiec, R. J.; Crabtree, R. H. Coordination chemistry of halocarbons. *Coord. Chem. Rev.* **1990**, *99*, 89–115.
- (48) Hanamoto, T.; Fuchikami, T. Remarkable dependency of diastereoselectivity on the selection of hydride source and Lewis acids in the reduction of 2-(trifluoromethyl)propiophenone. *J. Org. Chem.* **1990**, *55*, 4969–4971.
- (49) Mohanta, P. K.; Davis, T. A.; Gooch, J. R.; Flowers, R. A. I. Chelation-controlled diastereoselective reduction of α -Fluoroketones. *J. Am. Chem. Soc.* **2005**, *127*, 11896–11897.
- (50) Malamakal, R. M.; Hess, W. R.; Davis, T. A. Highly Efficient Diastereoselective Reduction of α -Fluoroimines. *Org. Lett.* **2010**, *12*, 2186–2189.
- (51) Stanton, G. R.; Norrby, P.-O.; Carroll, P. J.; Walsh, P. J. Chelation-Controlled Addition of Organozincs to α -Chloro Aldimines. *J. Am. Chem. Soc.* **2012**, *134*, 17599–17604.
- (52) Côté, A.; Boezio, A. A.; Charette, A. B. Catalytic asymmetric addition of diorganozinc reagents to N-phosphinoylalkylimines. *Proc. Natl. Acad. Sci. U. S. A.* **2004**, *101*, 5405–5410.
- (53) Wooten, A.; Carroll, P. J.; Maestri, A. G.; Walsh, P. J. Unprecedented Alkene Complex of Zinc(II): Structures and Bonding of Divinylzinc Complexes. *J. Am. Chem. Soc.* **2006**, *128*, 4624–4631.
- (54) Kim, H. Y.; Salvi, L.; Carroll, P. J.; Walsh, P. J. Highly Enantio- and Diastereoselective One-Pot Methods for the Synthesis of Halocyclopropyl Alcohols. *J. Am. Chem. Soc.* **2009**, *131*, 954–962.
- (55) Raffier, L.; Gutierrez, O.; Stanton, G. R.; Kozlowski, M. C.; Walsh, P. J. Alkenes as Chelating Groups in Diastereoselective Additions of Organometallics to Ketones. *Organometallics* **2014**, *33*, 5371–5377.